

Improved Delayed Neutron Group Parameters for Molten Salt Reactors

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Outline



Delayed neutron precursors (DNPs) emit delayed neutrons

Delayed neutron precursors (DNPs) are nuclides that undergo β^- decay, producing a daughter, called an emitter, which emits a delayed neutron [?].

Relevant data for DNPs includes:

$P_{n,i}$ = the emission probability [-]

τ_i = the half-life of DNP i [s]

$\chi_{d,i}$ = the delayed neutron energy spectrum from DNP i [-]

N_i = the number of atoms of i [-].

Accurately modeling the DNPs is important for transient simulations, which are important for safety, security, and safeguards by design.



DNP groups approximate DNPs

A small number of artificial DNPs, known as DNP groups, are used to approximate the DNPs to reduce computational cost.

The three parameters for DNP groups are:

$\bar{\nu}_{d,k}$ = the delayed neutron yield of DNP group k [-]

τ_k = the half-life of DNP group k [s]

$\chi_{d,k}$ = the delayed neutron energy spectrum of DNP group k [-]

Six of these DNP groups can accurately approximate the DNPs in a light water reactor [?].



Safety, security, and safeguards are important

Safety, security, and safeguards by design is the act of developing a facility with features and processes that inherently address safety, security, and safeguards requirements from the earliest stages of design.

- Safety - Protect the public and minimize accident risk [?].
- Security - Prevent actors from acquiring information, stealing nuclear material, or committing sabotage [?].
- Safeguards - Impede the diversion or undeclared production of nuclear material for weapons by a host state [?].

Safety, security, and safeguards by design requires accurate transient simulations.

Outline



An Image



Figure 1: A caption describing the image. [?].

A Table

Frames (slides) can have “blocks.”

This one is about a cat

A cat in a hat.

A cat

In a hat.



Columns



Sometimes things need to be put side by side, in two nice looking columns. Maybe one column involves a quotation.

Explicit is better than implicit. – The Zen of Python

And, also, perhaps, a logo.



Figure 2: A caption describing the image.
[?].



Some Code

I have to use the fragile syntax for code slides.

```
def meow(volume):  
    """Make a demanding noise at the specified volume  
  
    Parameters  
    -----  
  
    volume: int  
        The volume of the demand. No relation to importance.  
  
    Returns  
    -----  
  
    str  
        meow  
    """  
  
    o = 'o'*volume  
    return 'me'+ o + 'ow'
```

Cat Math: Part 1



$$x = y \tag{1}$$

where

$$x = \text{cats} \tag{2}$$

$$y = \text{peculiar} \tag{3}$$

Cat Math: Part 2

Everything in Beamer is just like in $\text{\LaTeX}$. Right down to the theorems.

Theorem (Pythagoras)

$$a^2 + b^2 = c^2$$

Corollary

$$x + y = y + x$$

Proof.

$$\omega + \phi = \epsilon$$



Outline





Blue and Orange are Fierce

Those are the Illini Colors. Use them like you see them in Figure ??.



Figure 3: Kristofer Hivju is pretty serious about this color palette [?].

Outline



Conclusion

We showed many things. This slide is an example of how you can animate bulleted lists, for more information about using beamer animations, checkout the overleaf article on overlay specifications in the group's guide.

- Cats are peculiar

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- Math can be rendered nicely

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- Cats are peculiar
- Blue and Orange are fierce colors
- Math can be rendered nicely
- Cite your sources

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